

Scaffold-redundant function

One Functional Specification S · many Murcko scaffolds · shared surrogate behavior

THE CLAIM

Behavior is scaffold-redundant under S :
chemically different cores
can realize the same Spec.

THIS NEIGHBORHOOD

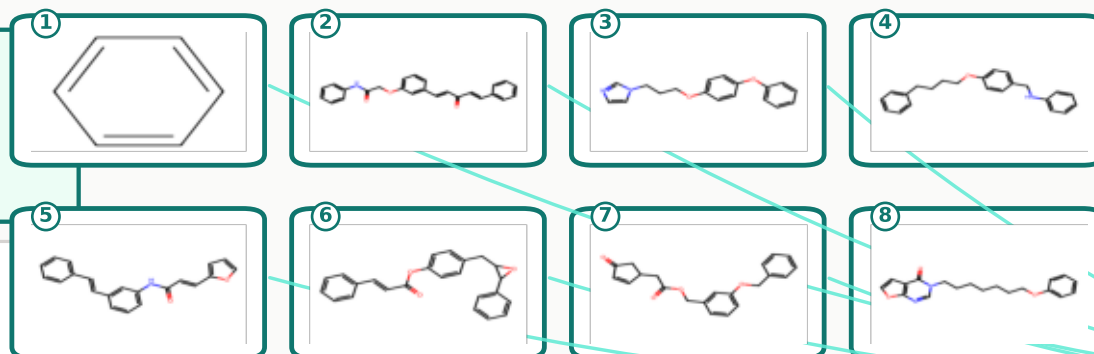
59

unique Murcko scaffolds
in S $k=64$ neighbors

GENERATION (G5)

114

scaffolds from one Spec
(batch LogP std=0.12)

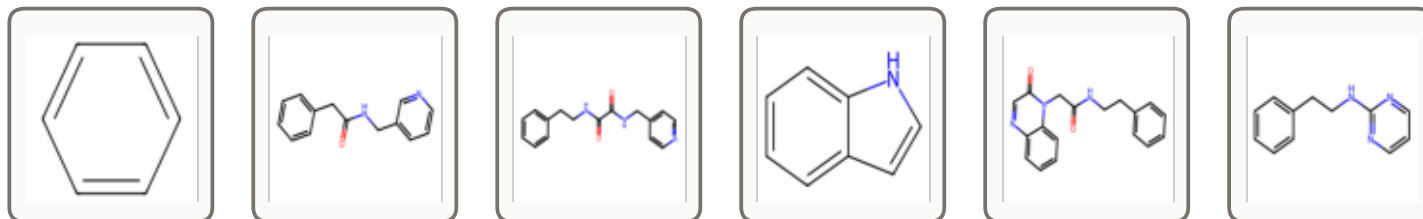


**ONE
SPEC S**
shared behavior

8 distinct Murcko cores → converge on the same Spec (scaffold redundancy)

Contrast: fingerprint neighbors for the same probe

ECFP $k=64$: 46 unique scaffolds · higher behavior variance ($v_{\text{beh}}=0.73$ vs $S=0.26$)



Structure-similar search does not target scaffold-redundant function

Same probe, two geometries

Unique scaffolds
(higher = more redundant)



Behavior variance
(lower = tighter function)



Scaffold entropy H
(higher = more diverse)



S
ECFP